Competitive Dynamic Pricing · Nash Equilibria

Multi-Agent RL · Oligopoly Simulation · Collusion Detection

WEEK 4 OF 8

PHASE 2 — RULE-BASED AGENTS → PHASE 3 TRANSITION

Week 4: Q-Learning Agent — Implement tabular Q-learning entirely from scratch this week — no Stable-Baselines3 yet. Students who don't hand-code the Bellman update at least once will treat DQN and PPO as black boxes forever. This is the most important implementation week of the project. Also the mid-project review — mentor must sign off before students proceed to deep RL in Week 5.

⌖ MID-PROJECT REVIEW — End of Week 4

This is a mandatory gate before Week 5. The Q-learning agent must consistently beat the Random baseline. If it doesn't, the environment or reward function likely has a bug — fixing it now saves the entire second half of the project.

Priority key: MUST WATCH / READ | RECOMMENDED | OPTIONAL | ■ = estimated time | pp. = page range

■ YouTube & Interactive Videos

Q-Learning Explained — A Reinforcement Learning Tutorial
deeplizard
youtube.com/watch?v=qhRNvCVVJaA
The best concise explanation of Q-tables, Bellman equations, and ε-greedy exploration online. Watch this before reading Sutton & Barto Ch. 6 — it makes the math significantly easier to absorb.
MUST WATCH ■ 19 min

Q-Learning: Model Free Reinforcement Learning and Temporal Difference
StatQuest with Josh Starmer
youtube.com/watch?v=AJiG9iFTdzU
StatQuest's signature clarity on temporal difference updates. The cleanest visual walkthrough of the Q-learning update rule available. Start here if the Bellman equation feels abstract.
MUST WATCH ■ 14 min

Bellman Equation Explained Clearly
Arxiv Insights
youtube.com/watch?v=14BfO5lMiuk
Builds the Bellman equation from scratch: Value Function → Q-Function progression. The mathematical heart of Q-learning — students must understand this before writing the update rule.
MUST WATCH ■ 12 min

Q-Learning — Full Python Implementation from Scratch
Nicholas Renotte
youtube.com/watch?v=YLa_KkehvGw
Full coding tutorial — students code along and then replace the grid world with their pricing environment. The single most practical resource this week. Code along, don't just watch.
MUST WATCH ■ 38 min

Exploration vs Exploitation — The ε-Greedy Strategy
deeplizard
youtube.com/watch?v=mo96Nqlo1L8
Covers ε-greedy policy, epsilon decay schedules, and why exploration matters. Students who set epsilon too low too fast end up with agents stuck in suboptimal strategies — this video explains exactly why.
MUST WATCH ■ 11 min

Hyperparameter Tuning for Reinforcement Learning
Machine Learning with Phil
youtube.com/watch?v=oX58QxoG7ms
Practical guidance on tuning learning rate α, discount factor γ, and epsilon decay. Useful once Q-learning is running but not converging well — a common Week 4 problem.
RECOMMENDED ■ 20 min

Reinforcement Learning Course — Q-Learning Deep Dive (Lecture 4)
Mutual Information (YouTube)
youtube.com/watch?v=AJiG9iFTdzU
University-grade lecture covering Q-learning convergence guarantees and the conditions under which the algorithm is proven to find the optimal policy.
RECOMMENDED ■ 35 min

Deep Q-Networks — Introduction (preview of Week 5)
Arxiv Insights
youtube.com/watch?v=wrBUkpiRvCA
Optional preview of what comes next in Week 5. Explains why tabular Q-learning breaks at scale and how neural networks fix it — useful context for motivated students.
OPTIONAL ■ 14 min

■ Papers, Books & Docs

■	**Reinforcement Learning: An Introduction — Ch. 6 (TD Learning & Q-Learning)** Sutton & Barto (Free PDF at incompleteideas.net) incompleteideas.net/book/RLbook2020.pdf The canonical source. Section 6.5 derives Q-learning from temporal-difference principles. Students must read §6.5 and implement the update rule on paper before coding.	**MUST READ** ■ ~50 min pp. 119–145 (focus §6.5)
■	**Q-Learning (original seminal paper)** Watkins & Dayan — Machine Learning Journal, 1992 doi.org/10.1007/BF00992698 Only 8 pages. The paper that introduced Q-learning. Reading it alongside S&B; Ch. 6 anchors the theory. Focus on the convergence guarantee — students are often surprised it exists.	**MUST READ** ■ ~25 min pp. 279–292 (8 pages)
■	**AI, Algorithmic Pricing and Collusion — Section 3 (re-read)** Calvano, Calzolari, Denicolò, Pastorello — QJE 2020 doi.org/10.1093/qje/qjz036 Re-read Section 3 now that students can code Q-learning. Match your agent configuration to theirs: state space design, action space, reward signal, and hyperparameter choices.	**MUST READ** ■ ~25 min pp. 13–22
■	**OpenAI Spinning Up — Introduction to RL (Q-Learning walkthrough)** OpenAI (Free online) spinningup.openai.com/en/latest/spinningup/rl_intro.html The cleanest reference implementation of Q-learning online. Interactive notebook format — good bridge between theory and the project's custom environment.	**RECOMMENDED** ■ ~30 min Web — spinningup.openai.com
✏	**Q-Learning with Python: Step-by-Step Tutorial** Towards Data Science towardsdatascience.com → search 'Q-Learning Python tutorial' Step-by-step implementation in a simple environment. Students replicate this first, then adapt the Q-table logic to the custom pricing environment.	**RECOMMENDED** ■ ~20 min Web article
■	**Reinforcement Learning: An Introduction — Ch. 3 (Finite MDPs)** Sutton & Barto incompleteideas.net/book/RLbook2020.pdf → Ch. 3 Formal Markov Decision Process framework. Optional but strongly recommended for students considering graduate research or who want to understand the theoretical foundation under Q-learning.	**OPTIONAL** ■ ~40 min pp. 47–71

■ Common Pitfalls to Watch — Week 4

■ Reward not scaled — profits can be in the hundreds; normalise to [0, 1] or Q-values will explode or vanish.

■ Epsilon decay too fast — start at ε = 1.0, decay slowly to 0.05 over 80% of training episodes. Premature exploitation is the #1 reason Q-learning fails to converge.

■ Q-table too large if the price grid is fine-grained — consider bucketing state-action pairs if memory becomes an issue.

■ Skipping the mid-project review — mentor must sign off at the end of Week 4 before the student proceeds to DQN/PPO in Week 5.

■ Videos (must)	■ Videos (rec.)	■ Reading (must)	■ Total (must only)	■ Total (all items)
94 min	55 min	100 min	~3.5 hr	~7 hr